Semester: 6th
Subject Name: -Cloud Computing& Code: - CS 30010
Branch (s): Computer Science and Engineering

SPRING **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Cloud Computing
[CS 30010]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

   a) What is the objective behind distributed data center?
   b) What is DRaaS and how to achieve DRaaS in cloud computing?
   c) What is elasticity in cloud computing and how it is different from scalability?
   d) What is data replication and how to achieve it?
   e) Specify the importance of hypervisor in a virtualized environment.

2. (A) Discuss any two scheduling techniques of your choice with example. [2.5 Marks]

   (B) Explain cloud computing architecture with a neat diagram and explain each component. [2.5 Marks]

3. You are migrating a VM from a source host to a destination host using live migration. The VM has a virtual disk that is stored on a shared storage device. During the migration process, the VM's memory is transferred to the destination host, but the virtual disk remains on the shared storage device. As the migration is completing, the VM on the source host is still writing data to the virtual disk. Meanwhile, the VM on the destination host is starting to boot up and access the same virtual disk. What happens to the data consistency of the virtual disk, and how do you ensure that the VM on the destination host sees a consistent view of the data? [5 Marks]

4. (A) Discuss cloud deployment models in detail. Differentiate hybrid cloud with multi cloud. [2.5 Marks]

   (B) Discuss in details any two techniques of virtual machine migration. [2.5 Marks]

5. (A) Write a short note on virtualization in cloud computing? [2.5 Marks]

   (B) What are the basic things need to consider before migrating towards cloud from conventional model. [2.5 Marks]

***** Best of Luck *****